



OPEN Capsule network approach for monkeypox (CAPSMON) detection and subclassification in medical imaging system

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In response to the pressing need for the detection of Monkeypox caused by the Monkeypox virus (MPXV), this study introduces the Enhanced Spatial-Awareness Capsule Network (ESACN), a Capsule Network architecture designed for the precise multi-class classification of dermatological images. Addressing the shortcomings of traditional Machine Learning and Deep Learning models, our ESACN model utilizes the dynamic routing and spatial hierarchy capabilities of CapsNets to differentiate complex patterns such as those seen in monkeypox, chickenpox, measles, and normal skin presentations. CapsNets' inherent ability to recognize and process crucial spatial relationships within images outperforms conventional CNNs, particularly in tasks that require the distinction of visually similar classes. Our model's superior performance, demonstrated through rigorous evaluation, exhibits significant improvements in accuracy, precision, recall, and F1 score, even with limited data. The results highlight the potential of ESACN as a reliable tool for enhancing diagnostic accuracy in medical settings. In our case study, the ESACN model was applied to a dataset comprising 659 images across four classes: 178 images of Monkeypox, 171 of Chickenpox, 80 of Measles, and 230 of Normal skin conditions. This case study underscores the model's effectiveness in real-world applications, providing robust and accurate classification that could greatly aid in early diagnosis and treatment planning in clinical environments.

Keywords Image classification, Multi-class classification, Deep learning, Capsule networks, Spatial awareness, Accuracy

Monkeypox, a zoonotic virus¹ emerging into the spotlight of global health concerns, traces its origins and molecular lineage to the Orthopoxvirus genus, sharing taxonomic kinship with variola (smallpox), vaccinia, and cowpox viruses. This connection underscores a significant aspect of its virology, ecology, and potential for human disease. Discovered in 1958 during an outbreak among monkeys kept for research, the first human case was not documented until 1970 in the Democratic Republic of Congo. Since then, monkeypox has not only remained a public health issue in central and western African countries but has also caused outbreaks in other parts of the world, indicating its potential for wider spread and impact. Statistically, the incidence of monkeypox has shown an alarming increase, especially in the last two decades. The disease, characterized by fever, rash, and swollen lymph nodes, can range from mild to severe, with fatalities more common among those with compromised immune systems.

The World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) have documented variable case fatality rates, which have historically ranged from 1 to 10% in different outbreaks. As observed from Fig. 1, recent data suggests a shift in the epidemiology of monkeypox, with increased human-to-human transmission compared to the traditionally understood zoonotic transmission route. This shift necessitates a deeper understanding of the virus's transmission dynamics, virulence factors, and the interplay between human activities and animal reservoirs.

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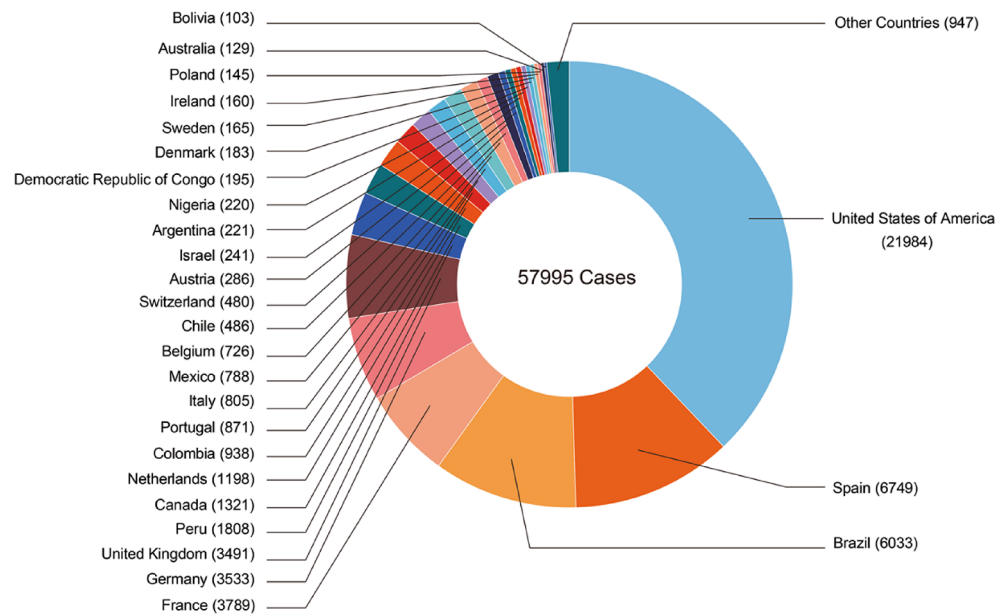


Fig. 1. Illustration of Monkeypox outbreak on a Global Scale².

The research gaps in the detection and classification of monkeypox, alongside similar infectious diseases such as chickenpox and measles, are multifaceted. Firstly, the clinical manifestations of these diseases can be highly similar, particularly in the early stages, making differential diagnosis challenging in the absence of specific laboratory tests. This challenge is compounded in resource-limited settings where access to advanced diagnostic facilities is scarce. Moreover, the global health community faces gaps in surveillance and reporting systems, which hampers accurate epidemiological tracking and response strategies. These gaps underscore the need for innovative approaches to disease detection and classification, where the integration of machine learning (ML) and deep learning (DL)³ models hold significant promise.

The motivation behind this research arises from the urgent need to enhance disease surveillance, diagnosis, and classification through the application of advanced computational technologies. ML and DL models, by virtue of their ability to learn from and make predictions based on large datasets, offer a transformative potential for the early detection and accurate classification of infectious diseases. These models can analyze diverse data types, including clinical symptoms, epidemiological patterns, and molecular sequences, to identify disease signatures that are indistinguishable to the human eye. The role of these computational models extends beyond diagnosis to include predictive modeling of disease spread, identification of high-risk areas, and informing public health interventions.

The significance of Capsule Networks (CapsNets)⁴ in such applications cannot be overstated. CapsNets, an evolution in the field of deep learning, provide a unique framework for image analysis that is particularly suited for medical imaging tasks. Unlike traditional convolutional neural networks (CNNs) that may lose spatial hierarchies between high-level features, CapsNets preserve these relationships, making them exceptionally adept at recognizing complex patterns in data. This capability is crucial for distinguishing between different infectious disease manifestations on the skin^{5–8}, such as the rashes caused by monkeypox, chickenpox, and measles. CapsNets can potentially offer a more accurate classification of these diseases, even in the presence of subtle or overlapping clinical features.

The increasing incidence and global spread of monkeypox underscore the urgent need for innovative approaches to its detection, classification, and management. This research aims to bridge existing gaps through the integration of ML and DL models, with a particular focus on the promising capabilities of Capsule Networks. By utilizing the power of these advanced computational tools, we can enhance our understanding of infectious diseases, improve diagnostic accuracy, and ultimately contribute to more effective public health responses. The integration of technology in disease surveillance and management not only signifies a leap forward in our fight against infectious diseases but also exemplifies the pivotal role of interdisciplinary collaboration in addressing complex global health challenges. The experimentation has been done in the dataset found in literature⁹.

The Enhanced Spatial-Awareness Capsule Network (ESACN) aims to address the critical limitations observed in traditional deep learning approaches, particularly Convolutional Neural Networks (CNNs), in dermatological disease detection and classification tasks. Traditional CNNs, while effective, suffer from inherent limitations, including their tendency to lose spatial hierarchies due to pooling operations and reliance on scalar outputs. This limitation is particularly problematic in dermatological imaging, where subtle variations in spatial orientation, texture, and lesion shape are critical for accurate diagnosis.

1. Dynamic Routing and Preservation of Spatial Hierarchies: The ESACN architecture integrates dynamic routing and capsule-based modeling to retain spatial relationships between features within images. Unlike

CNNs, which apply pooling to reduce dimensionality—resulting in the loss of spatial information—Capsule Networks preserve these hierarchies. For example, in distinguishing between Monkeypox and other visually similar conditions like Chickenpox or Measles, the ESACN model maintains the orientation, texture, and relative position of lesions, enabling a more nuanced understanding of the patterns critical for accurate classification.

2. **Data Efficiency and Robustness to Transformations:** The ESACN also improves on the efficiency of CNNs, especially in scenarios with limited data. Traditional CNNs require extensive data augmentation and large datasets to achieve high accuracy, but the ESACN's capsule structure leverages spatial hierarchies, reducing the need for vast amounts of annotated data. Moreover, its robustness to affine transformations—such as rotation, scaling, and translation—ensures accurate performance even when image orientations vary, a common challenge in real-world clinical datasets.
3. **Interpretability and Model Transparency:** Finally, the ESACN model enhances interpretability, an essential aspect for clinical application. While CNNs often act as “black-box” models, CapsNets provide a structured approach where the relationships between features are explicitly modeled and traceable. This transparency is crucial in medical diagnostics, as clinicians must understand the basis of a model's decision. For instance, the ESACN not only classifies skin conditions but also explains how specific spatial relationships between features contribute to the classification, thereby increasing trust and reliability in clinical settings.

By addressing these limitations, the ESACN model demonstrates a significant advancement over conventional CNN architectures, proving effective in multi-class classification tasks and improving the diagnostic accuracy necessary for dermatological disease management.

Related work

Literature¹⁰ primarily focuses on Machine Learning (ML) and Deep Learning (DL) applications for detecting and diagnosing mpox, alongside evaluating public sentiment regarding the disease and aiding in drug repurposing efforts. Techniques such as BERT for NLP and LSTM for forecasting are highlighted, with models evaluated on metrics like mean absolute error (MAE) and R-squared. Common limitations across the studies include an insufficiency of open-source data for training robust AI models and the challenges associated with managing data privacy. Additionally, the research highlights the necessity for model adaptability to account for viral mutations and geographical diversity. The reviewed literature evidences a significant reliance on skin lesion imagery for diagnosing mpox, suggesting a potential bias towards this mode of detection. In¹¹ a deep learning approach using Convolutional Neural Networks (CNNs) enhanced by Grey Wolf Optimizer (GWO) algorithm is proposed to classify monkeypox skin lesions. This study handles the challenge of accurate diagnosis of monkeypox by visual inspection of lesions, particularly in resource-limited settings. The optimized CNN model exhibits a significant improvement in classification performance metrics such as accuracy (95.3%), precision, recall, F1-score, and AUC, compared to its non-optimized counterpart. The GWO optimization strategy notably enhances the CNN's discrimination capability between positive and negative cases. The approach demonstrates potential benefits in early detection and treatment of monkeypox, with wider public health implications for controlling outbreaks. The study also projects the need for further research in data collection, algorithm generalizability, and integration with healthcare systems¹². investigates an advanced Deep Learning (DL) method for identifying Monkeypox in skin lesion images, evaluating models on a dataset of Monkeypox, Chickenpox, Measles, and normal cases. Seven Deep Neural Network (DNN) models have been presented with two-class and four-class detection scenarios. The DenseNet201-based model outperformed other models with a two-class scenario accuracy of 97.63%, F1-Score of 90.51%, and AUC of 94.27%; and a four-class scenario accuracy of 95.18%, F1-Score of 89.61%, and AUC of 92.06%. The study sets itself apart by providing model explainability using LIME and Grad-CAM thereby enhancing the understanding of AI-driven diagnostics and bolstering clinicians' trust in these models. The need for comprehensive datasets and robust models adaptable to various clinical features for real-world applicability is highlighted.

Literature¹³ presents a cutting-edge approach utilizing explainable AI (xAI) to enhance the effectiveness of convolutional neural networks (CNNs) in classifying Monkeypox skin lesions. Utilizing a dataset of 572 images divided into Monkeypox and Normal classes, the study evaluates 12 different CNN models, with MobileNet V2 emerging as the most accurate, achieving 98.25% accuracy, 96.55% sensitivity, 100% specificity, and a 98.25% F1-Score. The integration of explainable AI methods with the MobileNet V2 model projects the importance of transparency in AI-driven diagnostics, providing insights into the decision-making process of the model. This approach not only elevates the accuracy of Monkeypox detection but also significantly contributes to the development of auxiliary diagnostic systems in healthcare, potentially alleviating hospital congestion during outbreaks¹⁴. presents a novel application aimed at enhancing the speed and accuracy of Monkeypox detection. Utilizing a deep learning framework with a MobileNetV2 model, the study achieves a remarkable 91.11% accuracy in classifying skin lesions as Monkeypox-infected or not. However, the study acknowledges potential limitations such as the variability in image quality and the general challenge of gathering a comprehensive dataset that accurately represents the diverse manifestations of Monkeypox lesions across different populations¹⁵. investigates the application of transfer learning and meta-heuristic optimization for enhancing the classification accuracy of monkeypox images. Utilizing the GoogleNet deep network for feature extraction, the study introduces two novel algorithms: the AI-Biruni Earth Radius (BER) optimization algorithm and a hybrid optimization method combining the strengths of the Sine Cosine Algorithm (SCA) and Particle Swarm Optimization (PSO). Tested on a dataset comprising monkeypox and normal skin images, the proposed methods demonstrated a significant improvement in classification performance, achieving an average accuracy of 98.8%. Despite its promising results, the study acknowledges the need for further validation on larger and more diverse datasets¹⁶. presents a novel approach for diagnosing Monkeypox through skin images using a hybrid artificial intelligence

system. This work combines state-of-the-art deep learning models with long short-term memory (LSTM) to improve classification accuracy in a multi-class structure consisting of chickenpox, measles, monkeypox, and normal classes. The study addresses the challenge of unbalanced data through extensive data augmentation and preprocessing, achieving a test accuracy of 87% and a Cohen's kappa score of 0.8222. It also emphasizes the use of convolutional neural networks (CNN), VGG-19, Inception V3, ResNet-50, and Autoencoder models, detailing their architecture, output shapes, and the number of parameters. The research demonstrates the potential of these models in accurately identifying Monkeypox through skin lesion images, contributing significantly to early detection efforts. The study utilizes a diverse dataset, applying preprocessing techniques to enhance model performance and using metrics such as accuracy, sensitivity, specificity, and F1-score for evaluation. Merits of the study include the application of advanced deep learning models to a critical health concern and the provision of a detailed model structure that can aid in the replication of results. Limitations include the need for extensive computational resources due to the complex model architectures and the reliance on high-quality, annotated images for training. In¹⁷, a patch-based Vision Transformer (ViT) model combined with transfer learning is introduced for identifying monkeypox and chickenpox diseases from skin color images. The innovative approach utilizes a 16×16 patch size for the images, enhancing the model's ability to discern subtle dermatological features specific to each disease, leading to an accuracy, precision, recall, and F1 score of 93%. This method signifies a substantial leap forward in medical diagnostics, utilizing deep learning's power to improve early disease detection accuracy. Despite its high accuracy, the study acknowledges the ongoing challenges of data scarcity and the need for models that can adapt to diverse datasets.

By integrating Convolutional Neural Networks (CNN) for segmentation and RootSIFT for classification enhancement¹⁸, advances the precision in identifying and classifying monkeypox lesions. The methodology involves multi-task learning (MTL) that simultaneously addresses lesion segmentation and classification, utilizing shared representations for both tasks to improve overall accuracy and efficiency. Experimental results demonstrate the method's superiority over traditional techniques, achieving improved classification accuracy. This research highlights the potential of combining advanced image analysis methods for accurate and efficient skin lesion analysis, which could significantly aid dermatologists in precise diagnosis and categorization of skin diseases. By fine-tuning the hyper-parameters of Yolov5 enhanced with a convolutional neural networks and transfer learning techniques¹⁹, achieves a classification accuracy of 98.18% on monkeypox skin lesion images. The utilization of hyperparameter optimization strategies such as SGD optimizer, Bayesian optimizer, and Learning without Forgetting (LwF) significantly boosts the model's performance. Despite its high accuracy, the study acknowledges challenges like the scarcity of open-source datasets for comprehensive training and the generalizability of models across diverse population groups and conditions. Utilizing deep learning models like InceptionV3, EfficientNet, VGG16, and integrating Squeeze and Excitation Network (SENet) as an attention model²⁰, marks the first to apply SENet for Monkeypox classification. The innovation does not stop with model architecture; the study also integrates datasets from different sources, enhancing model performance significantly across accuracy, precision, recall, and F1-score metrics. Despite achieving substantial results, the study acknowledges the ongoing challenge of data scarcity.

Utilizing data from Global Health and the World Health Organization²¹, analyzes symptoms' relationships with monkeypox through gradient boosting methods like Extreme Gradient Boosting (XGBoost), CatBoost, LightGBM, alongside Support Vector Machine (SVM) and Random Forest. XGBoost outperformed other methods with perfect accuracy in tests, validated through k-fold cross-validation averaging 90% accuracy. Shapley Additive Explanations (SHAP) further interpreted the XGBoost model's output, emphasizing the importance of symptoms like fever, skin lesions, and headache in diagnosis. This research introduces a novel, symptom-based approach to diagnosing monkeypox, differing from previous image-based studies, and highlights the effectiveness of ML models in disease detection based on symptomatic data²². utilizes watershed segmentation along with a convolutional neural network (CNN)-based autoencoder to accurately differentiate Monkeypox skin lesions from other conditions. Watershed segmentation helps in isolating individual lesions in images, while the CNN-based autoencoder efficiently processes these images to highlight features relevant to Monkeypox. The combination of these techniques addresses the challenge of similar appearances between Monkeypox lesions and other dermatological conditions, aiming for high accuracy and quick diagnosis. The reliance on high-quality images and the computational demand of processing large datasets are potential limitations. Employing models like MobileNetV3-s, EfficientNetV2, and ResNET50 among others^{23–25}, utilizes a customized hybrid transfer learning model to achieve remarkable classification accuracy. With an average F1-score of 0.98, AUC of 0.99, accuracy of 0.96, and recall of 0.97, the optimized hybrid MobileNetV3-s model outperformed other configurations. The research stands out by integrating hyperparameter optimization and model customization to handle the monkeypox classification challenge effectively, offering a significant contribution to automated disease detection. However, the paper also acknowledges the limitation of relying on available datasets, suggesting further exploration and model combination for enhanced outcomes²⁶. uses pre-trained convolutional neural networks (CNNs) like VGG16, VGG19, ResNet50, ResNet101, DenseNet201, and AlexNet. The study combines these CNNs with an adaptive k-means clustering image segmentation technique, aiming for precise and straightforward segmentation outcomes. It also introduces Local Interpretable Model-Agnostic Explanations (LIME) for a deeper understanding of the model's feature extraction and classification decisions. The best accuracy achieved by ResNet101 was 94.25% with an AUC of 98.59%, highlighting the potential of combining deep learning with image segmentation and interpretability techniques for medical image analysis, specifically for monkeypox detection. This approach seeks to offer more accurate and quicker diagnoses compared to manual image analysis, despite potential challenges related to image quality and computational demand.

A novel fully automatic technique for brain tumor regions segmentation proposed in²⁷ by using multiscale residual attention-UNet (MRA-UNet). MRA-UNet uses three consecutive slices as input to preserve the

sequential information and employs multiscale learning in a cascade fashion, enabling it to exploit the adaptive region of interest scheme to segment enhanced and core tumor regions accurately. A significant improvement has been demonstrated for the core and enhancing tumor region segmentation showing the proposed method's evident effectiveness. A multi-task semi-supervised learning (MTSSL) framework is proposed in²⁸ that utilized auxiliary tasks for which adequate data is publicly available. The generalization of framework is improved due to this semi-supervised learning and thus it leads to enhancement in COVID-19 detection performance. The proposed model is rigorously evaluated on the largest publicly available COVID-19 dataset and experimental results show that the proposed model attained state-of-the-art performance.

A densely attention mechanism-based network (DAM-Net) for COVID-19 detection is proposed by²⁹ in CXR. DAM-Net adaptively extracts spatial features of COVID-19 from the infected regions with various appearances and scales. Experimental results demonstrate that this approach obtains state-of-the-art results for COVID-19 classification with an accuracy of 97.22%, a sensitivity of 96.87%, a specificity of 99.12%, and a precision of 95.54%.

Limitations in existing works

The limitations observed in the discussed literature works primarily revolve around the dependence on large, high-quality datasets, the computational intensity required for training deep convolutional neural networks (CNNs), and the potential for overfitting due to the sophisticated architecture of pre-trained models. Additionally, while models like ResNet101 achieved high accuracy, they inherently lack interpretability, making it difficult for medical professionals to understand the basis of the model's decisions. The use of Local Interpretable Model-Agnostic Explanations (LIME) aims to address this, yet it still represents a workaround rather than an intrinsic solution to the model's transparency. Additionally, a significant limitation in the existing literature is the focus on single-class or binary classification models, which may not be sufficient for complex medical diagnostics where multiple conditions can present with similar symptoms. This is particularly relevant in the context of skin diseases, where distinguishing between conditions like monkeypox, chickenpox, measles, and normal skin requires precise differentiation based on subtle image features.

The above limitations highlight the need for a Capsule Network (CapsNet) model in the proposed work for several reasons. CapsNet aims to understand the spatial hierarchies between features, potentially reducing the need for extensive data augmentation and large datasets to achieve high performance. By focusing on the hierarchical relationships between parts and wholes in an image, CapsNets can potentially reduce the computational overhead required for training, as they aim to learn more with less data. CapsNets, through their dynamic routing and capsule architecture, provide a more interpretable framework by modeling the spatial relationships and poses of objects in the image. This could offer clearer insights into the decision-making process, aligning with the medical field's need for understandable diagnostic tools. Unlike traditional CNNs, CapsNets are designed to be more robust to spatial variations in the data, such as rotation, translation, and other affine transformations. This characteristic is crucial for medical image analysis, where the orientation and position of diagnostic features (e.g., skin lesions in monkeypox detection) can vary significantly across images.

In summary, the proposed CapsNet model addresses the limitations of traditional CNN approaches by offering a more data-efficient, computationally viable, and inherently interpretable method for medical image analysis, specifically for the detection and classification of diseases like monkeypox. This makes it a promising direction for enhancing the accuracy and usability of automated diagnostic systems in healthcare.

The primary motivation for this study arises from the need to improve the accuracy and reliability of dermatological disease classification, particularly for conditions such as monkeypox, chickenpox, measles, and normal skin presentations. Traditional Machine Learning (ML) and Deep Learning (DL) models, such as Convolutional Neural Networks (CNNs), have shown limitations in distinguishing between these visually similar conditions due to their reliance on scalar outputs and pooling operations that can lose spatial hierarchies and relationships within images. Our Enhanced Spatial-Awareness Capsule Network (ESACN) addresses these limitations by utilizing the dynamic routing and spatial hierarchy capabilities of Capsule Networks (CapsNets). CapsNets preserve the spatial relationships between features in an image, making them particularly adept at recognizing and differentiating complex patterns that are crucial in medical imaging tasks. This inherent ability of CapsNets to process spatial hierarchies allows our ESACN model to achieve superior performance in multi-class classification tasks, which is essential for accurate diagnosis and treatment planning. Additionally, our work addresses the challenge of limited data availability, a common issue in medical imaging. The ESACN model demonstrates significant improvements in accuracy, precision, recall, and F1 score, even with a relatively small dataset. This robustness to data scarcity is crucial for practical applications in clinical settings, where, large, annotated datasets may not always be available. By improving diagnostic accuracy and reliability, our ESACN model has the potential to enhance early detection and treatment planning for dermatological diseases, ultimately contributing to better patient outcomes.

Dermatological datasets, particularly for diseases such as Monkeypox, Chickenpox, and Measles, present several challenges that were encountered during this study:

- **Limited availability of publicly accessible data:**
 - Monkeypox: Since Monkeypox is relatively rare and has seen a resurgence only recently, there are limited publicly accessible and annotated datasets. Most existing images are dispersed across various platforms, often lacking consistent clinical validation. This scarcity necessitated extensive searches across multiple medical repositories and online databases to gather sufficient and reliable images.
 - Chickenpox and Measles: While more common than Monkeypox, datasets for Chickenpox and Measles are often focused on pediatric cases and may not cover the full spectrum of lesion presentations. Furthermore,

many images are not publicly available due to privacy regulations surrounding pediatric dermatological data.

• **Quality and consistency of images:**

One significant challenge was ensuring that all images met a minimum quality standard for resolution and clarity. Many available images vary in terms of lighting conditions, angles, and focus, which can affect the model's ability to accurately classify skin conditions. We had to filter through numerous sources to select images that maintained consistent quality, ensuring the model was trained on reliable data.

Additionally, consistency in labeling was a concern, as not all images come with verified clinical annotations. Ensuring the accuracy of labels required cross-referencing multiple sources and, in some cases, expert consultation to validate the authenticity of the diagnostic information provided.

• **Diverse skin types and demographics:**

The diversity of skin types and demographics represented in existing datasets is another challenge. Many publicly available datasets predominantly feature lighter skin tones, which can bias model training and reduce generalizability. To mitigate this, efforts were made to collect images from diverse sources that cover a broader range of skin tones. However, achieving an equitable representation remains an ongoing challenge due to the limited availability of diverse data.

• **Balancing dataset size across classes:**

Ensuring a balanced dataset across the four classes (Monkeypox, Chickenpox, Measles, and Normal skin) proved difficult due to the disparity in the number of available images for each condition. While Normal skin images were more readily available, cases for Monkeypox, Chickenpox, and Measles were relatively scarce, leading to an imbalance. Data augmentation techniques were employed to address this issue, but it is acknowledged that these methods cannot fully compensate for the inherent limitations in dataset availability.

Proposed work

A. *Basic Architecture of CapsNet.*

Capsule Networks represent a cutting-edge technology in deep learning. They are especially suited for tasks that benefit from the hierarchical structure of data. A capsule is a group of neurons whose outputs represent different properties of the same entity. Unlike neurons in a CNN that scalar outputs, capsules output vectors where the orientation represents the properties of the feature, and the length (or norm) indicates the probability of the feature's existence. The overall block schematic of proposed work is depicted in Fig. 2.

A basic illustration of CapsNet architecture is depicted in Fig. 3.

The basic architecture consists of primary capsules and digit capsules followed by dynamic routing mechanism.

An important parameter with respect to CapsNet architecture is the squashing function $S(\cdot)$ and computed as

$$S(\cdot) = \frac{\|v_j\|^2}{1 + \|v_j\|^2} \frac{v_j}{\|v_j\|} \tag{1}$$

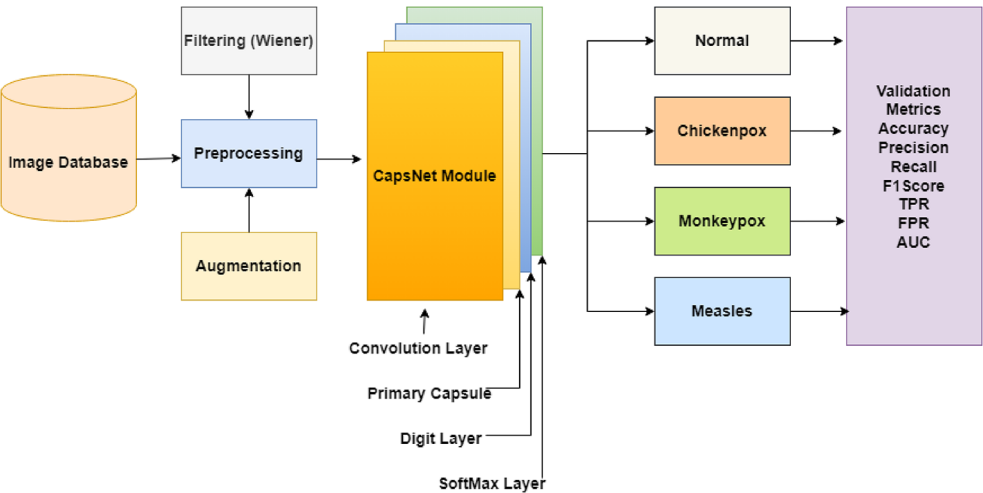


Fig. 2. Illustration of proposed multi class classification model.

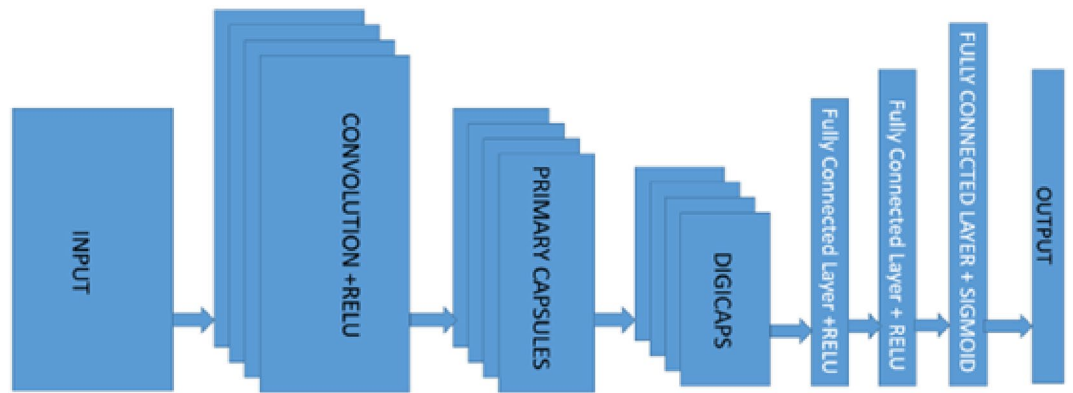


Fig. 3. Illustration of a general CapsNet architecture⁴.

In (1), v_j denotes the output vector of capsule j ; $\|\cdot\|$ denotes the Euclidean norm function. $S(\cdot)$ ensures that the length of the output vector is between 0 and 1. The squashing function ensures that the length of each capsule's output vector remains between 0 and 1, where the length (or magnitude) indicates the probability of the presence of a feature. This function maintains the orientation of the vector, which encodes important information about the detected feature's properties, such as its rotation or scale. The purpose of the squashing function is to normalize the output of each capsule so that it fits within a range suitable for the subsequent layers. This prevents the network from producing overly large output values, ensuring stability during training.

The primary capsules are the first layer of capsules that perform the initial high-dimensional entity detection in an image. Primary capsules form the foundational layer that directly interfaces with the raw features extracted by the initial convolutional layers. These primary capsules take the scalar outputs from the convolutional layers and transform them into vector outputs that represent the instantiation parameters of various features. The instantiation parameters can include position, size, orientation, and other attributes that could define the state of the feature being detected. The number of primary capsules is typically chosen based on the complexity of the features within the dataset and the desired level of feature representation. For simple datasets, fewer primary capsules are sufficient, while complex datasets with complex spatial hierarchies may require a larger number of capsules to capture the relationships adequately. The choice of the number of primary capsules is often determined empirically through experimentation and is guided by the performance of the network on a validation set. They take feature maps produced by traditional convolutional layers and transform them into small vectors which carry more specialized information.

The prediction vector U_{ij} by capsule i for capsule j is computed as

$$U_{ij} = W_{ij}u_i \quad (2)$$

In (2), W_{ij} denotes the weight matrix between capsules i and j ; u_i denotes the output of capsule i . This equation describes how the capsules in one layer interact with the capsules in the next layer. The weight matrix W_{ij} essentially determines how information is passed and modified between layers, learning the relationships between simpler and more complex features in the images.

Higher-level digit capsules represent more complex entities and are achieved by dynamic routing between capsules. They take predictions U_{ij} as inputs and use the routing by agreement mechanism to produce final outputs.

The degree of influence that one capsule should have on another capsule is computed in an iterative manner given by

$$c_{ij} = \frac{\exp(b_{ij})}{\sum_k \exp(b_{ik})} \quad (3)$$

where b_{ij} are the initial logits, or the log prior probabilities that capsule i should be coupled to capsule j , and k is the number of possible parent capsules in the layer above. Dynamic routing allows the network to adjust these connections based on the agreement between capsules. This process ensures that capsules with aligned outputs (indicating they are likely detecting related features) strengthen their connection, helping the model to build a more coherent representation of the input image.

The total input to a capsule s_j is then a weighted sum over all prediction vectors:

$$s_{ij} = \sum_k c_{ik}U_{ik} \quad (4)$$

and the output of capsule j is simply the squashed total input:

$$v_j = S(s_j) \quad (5)$$

By summing the inputs from lower-level capsules, the network combines the information gathered from different parts of the image, enabling higher-level capsules to form more complex and accurate feature representations. The output of (5) is a vector whose length indicates how likely the feature is present, while its orientation encodes additional properties (e.g., texture or orientation), maintaining the essential spatial details of the feature.

The final stage of the CapsNet is the dynamic routing process. It enables the network to focus on the most relevant parts of the input. The dynamic routing mechanism is the process that allows a capsule to decide to which parent capsule in the next layer it sends its output. Unlike traditional CNNs where the pooling layer forces the selection of dominant features, dynamic routing allows capsules to make a more precise decision based on the input context. The functionality of the dynamic routing mechanism is based on the “routing by agreement” concept. If the prediction made by a lower-level capsule agrees with the higher-level capsule’s output, then its input is considered more relevant for that higher-level capsule’s final output. The coupling coefficients between capsules, which determine how much a lower-level capsule’s output should be routed to a higher-level capsule, are updated iteratively through a “routing SoftMax” whose initial logits are the dot products of lower-level capsule outputs and higher-level capsule predictions.

This iterative process ensures that the network learns a part-whole relationship between features and entities in a way that accounts for the spatial orientation and transformation of features within the input space. This is particularly crucial for medical image analysis, where the spatial arrangement and structure of features can be indicative of the presence and severity of a condition like Monkeypox. Dynamic routing allows CapsNets to be more equivariant to spatial transformations, meaning that if the spatial relationships in the input image change, the output of the capsule network will change in a predictable way, thus preserving spatial hierarchies and relationships in the data.

The general algorithm is given as.

Algorithm for Dynamic Routing
Start with all $b_{ij} = 0$
For each iteration r
Compute coupling coefficient c_{ij} from b_{ij}
Compute weighted sum of total inputs s_{ij}
Compute $v_j = S(s_j)$
Measure the agreement factor as $a_{ij} = v_{ij}U_{ij}$
Update $b_{ij} \leftarrow b_{ij} + a_{ij}$

The process of dynamic routing by agreement iteratively refines the connections between capsules to ensure that the network becomes more sensitive to the spatial hierarchies and features within the data. By minimizing the loss that is partly defined by the discrepancy between the predicted vectors and the actual instantiation parameters, the network learns to recognize features and their spatial relationships effectively.

B. Proposed Model: Enhanced Spatial-Awareness Capsule Network (ESACN).

The Enhanced Spatial-Awareness Capsule Network (ESACN) is a novel architectural evolution of the standard Capsule Network (CapsNet) that incorporates an innovative spatial-awareness mechanism. The ESACN as shown in Fig. 4 is designed to improve the detection and classification of complex medical images, such as those used in identifying Monkeypox infections, by enhancing the network’s ability to recognize and encode the spatial relationships between different features in the images. The proposed ESACN model consists of multiple layers of capsules with a specialized spatial-awareness module that is integrated into the dynamic routing process. This model addresses the challenge of encoding spatial hierarchies by introducing additional capsules, called Spatial Capsules, which are specifically tasked with encoding and preserving the spatial orientation and location information of features within an image.

The novelty in proposed work lies in the spatial-awareness mechanism within ESACN model. This mechanism involves a novel set of Spatial Capsules that are interposed between the Primary Capsules and Digit Capsules layers. Each Spatial Capsule is responsible for capturing and encoding explicit spatial information about features which are quite crucial such as their orientation, scale, and position within the image.

The first layer in the proposed ESACN model is the convolution and pooling layer. Considering the image dataset $D = \{I_1, I_2, I_3, \dots, I_n\}$ where I_i depicts the individual images in the dataset and n is the total number of images in the dataset, each image in the dataset could be represented in terms of pixel intensities in the form of a matrix as $I_i(x, y, c)$ where (x, y) denote the spatial locations, and $c \in \{R, G, B\}$ denotes the color channel.

The initial step involves feature extraction which utilizes a single convolutional layer with a set of filter kernels $\{F_1, F_2, \dots, F_m\}$ where m denotes the number of filters in the layer. The output of the convolution layer is a feature map reflected as

$$M_{i,k}(x,y) = (F_k * I_i)(x,y) = \sum_{a=-A}^A \sum_{b=-B}^B \sum_c F_k(a,b,c) \cdot I_i(x+a,y+b,c) \tag{6}$$

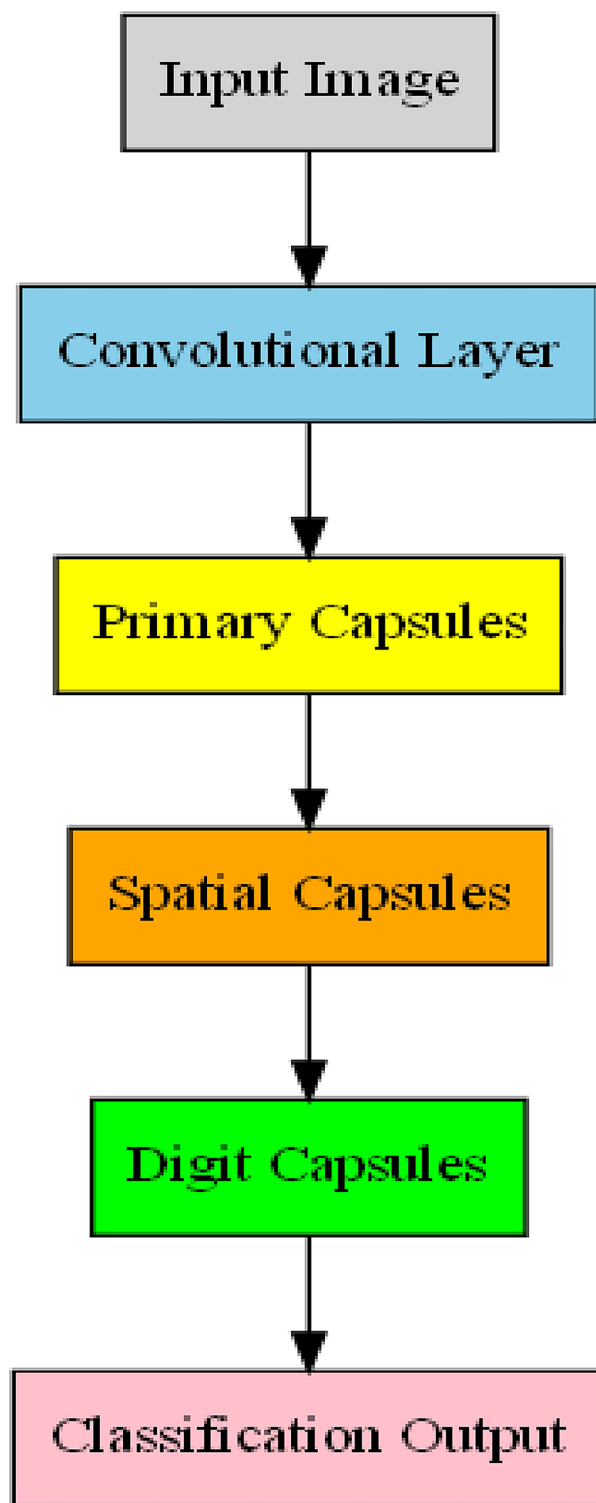


Fig. 4. Architectural layout of proposed ESCAN model.

The sums over a and b iterate over the spatial dimensions of the filter, and the sum over c iterates over the color channels.

In order to introduce non-linearity, a non – linear activation function is applied to the output of (6)

$$A_{i,k}(x,y) = \varnothing M_{i,k}(x,y) \quad (7)$$

In (7), ReLU is used as the activation function and denoted as $\varnothing$. $A_{i,k}(x,y)$ denotes the activated feature map.

In order to reduce the spatial dimension, a pooling layer is introduced to (7) to provide a spatial invariance as

$$P_{i,k}(x', y') = \text{Pool}(A_{i,k}(x, y)) \quad (8)$$

The initial feature extraction process transforms raw images I_i into a set of activated and pooled feature maps $\{P_{i,1}, P_{i,2}, \dots, P_{i,m}\}$ which serve as the input to the subsequent primary capsules layer. This process captures essential visual features from the images, laying the foundation for the more complex spatial and feature hierarchy processing that follows in the ESACN model.

Following the pooled output in (8), the next step involves transforming these feature maps into a set of primary capsule outputs. Each primary capsule receives inputs from a localized region of the pooled feature maps and produces a vector output that represents the instantiation parameters of a particular feature. The translation of pooled feature map into vector form is represented as

$$V_{ij} = \text{CapsTransform}(P_{i,k}(x', y'), W_j) \quad (9)$$

In (9) V_{ij} represents the output vector of the j -th primary capsule for image I_i , W_j denotes the weights associated with j -th primary capsule. Their primary objective is to learn to capture specific information in vector format. The transformation function typically involves a weighted sum over the inputs followed by a non-linear “squashing” function to ensure the output vector’s length represents the probability of the feature’s presence, while its orientation encodes the feature’s properties. It is denoted as

$$V_{ij} = \text{squash} \left(\sum_k (W_{jk} \cdot P_{ik}) \right) \quad (10)$$

In (10), the squashing function is defined as in (1).

Each primary capsule outputs a vector V_{ij} where the length of the vector indicates the probability that the feature it represents exists in the image, and its orientation represents the instantiation parameters (e.g., pose, texture, deformation) of the feature. The set of vectors $\{V_{i,1}, V_{i,2}, \dots, V_{i,J}\}$ for all primary capsules j form the input to the next layer of the capsule network. This layer consists of more sophisticated capsules capable of recognizing more complex patterns or objects by considering the outputs of multiple primary capsules together. This is where dynamic routing mechanisms come into play by guiding the interaction between capsules across layers to form a hierarchy of features and objects represented in the network.

For each primary capsule output V_{ij} spatial information is encoded into a separate vector S_{ij} , capturing spatial attributes like relative positioning, orientation, etc. from

$$S_{ij} = \text{SpatialEncode}(I_i) \quad (11)$$

Following this, the enhanced prediction vectors are generated as

$$U_{i,jl}^1 = W_{jl}^f V_{ij} + W_{jl}^s S_{ij} \quad (12)$$

In (12), W_{jl}^f and W_{jl}^s denote the weight matrices for feature and spatial information.

The objective at this stage is to compute the input to each digit capsule denoted as $D_{i,l}$ for the l -th digit capsule and image I_i , where there are L digit capsules in total. The modified dynamic routing algorithm is given below.

Modified Algorithm for Dynamic Routing
Start with all $b_{ij} = 0$
For each iteration r
Compute coupling coefficient c_{ijl} from b_{ij} using softmax as
$c_{ijl} = \frac{\exp(b_{ijl})}{\sum_k \exp(b_{ikl})}$
Compute weighted sum of total inputs $s_{ijl} = c_{ijl} U_{i,jl}^1$
Apply Squashing Function $D_{i,l} = \text{Squash}(S_{i,l})$
Update $b_{ijl} \leftarrow b_{ijl} + U_{i,jl}^1 D_{i,l}$

Finally, length of digit capsule output vectors $D_{i,l}$ is used to determine the presence and properties of higher-level features or classifications for image I_i . The class of image I_i can be determined by identifying the digit capsule with the maximum vector length:

$$\text{class}(I_i) = \text{argmax}_i \|D_{i,l}\| \quad (13)$$

The above sequence of mathematical formulations integrates spatial awareness directly into the capsule network’s processing flow. This enhances the model’s ability to learn complex patterns and relationships crucial for tasks such as disease detection. The ESACN architecture utilizes both the feature information captured by primary

capsules and the critical spatial context, processed through enhanced dynamic routing, to achieve superior performance in recognizing and classifying medical images.

The overall algorithm is given below.

Algorithm: Enhanced Spatial-Awareness Capsule Network (ESACN) for Image Classification
Input: Image dataset $I() = \{I_1, I_2, \dots, I_n\}$, where each I_i represents an individual image.
Output: Classification Labels $L() = \{L_1, L_2, \dots, L_n\}$
<i>Begin:</i>
For each image I_i in $I()$:
$F() = \{Conv(I_i)\}$
$P() = \{Squash(W_p * F() + B_p)\}$
For each pixel (x, y) in I_i :
$S_i(x, y) = Sum(W_s * P(x, y))$
Initialize routing coefficients $b_{ij} = 0$
For r iterations:
$C_{ij}(x, y) = Softmax(b_{ij} + I_i(x, y) \cdot U_{ij})$
$S_j = Sum(C_{ij} * U_{ij})$ // U_{ij} is the prediction vector
$V_j = Squash(S_j)$ // V_j are the Digit Capsule activations
If $r < routing_iterations$:
Update $b_{ij} += DotProduct(V_j, U_{ij})$
$I_{i_recon} = Decoder(V_j)$
$L_{class} = MarginLoss(T, V_j)$
$L_{recon} = Sum((I_i - I_{i_recon})^2)$
$L_{spatial} = Sum(Grad(I_i) - Grad(I_{i_recon}))$
$TotalLoss = L_{class} + \alpha * L_{recon} + \beta * L_{spatial}$
$L_i = ArgMax(V_j)$
Return $L() = \{L_1, L_2, \dots, L_n\}$
<i>End</i>

Results and discussion

Dataset

The recent increase in monkeypox cases has raised alarms among global healthcare communities, highlighting the importance of prompt identification to counter its swift spread. In response, a novel dataset focused on skin images for monkeypox detection has been compiled, featuring four distinct categories: Monkeypox, Chickenpox, Measles, and Normal. Sourced entirely from online platforms, this dataset was meticulously assembled by the Department of Computer Science and Engineering at Islamic University, Kushtia-7003, Bangladesh [Diponkar Bala, 2022]. Sample images from the four classes are depicted below from Figs. 5, 6, 7 and 8. In this study, a total of 659 images were used, distributed across four classes as follows: 178 images for Monkeypox, 171 images for Chickenpox, 80 images for Measles, and 230 images for Normal skin conditions as shown in Table 1. Figure 9 represents the class distribution of original data set.

Source and selection criteria To ensure the dataset was diverse and representative, the following criteria were applied during the image selection process.



Fig. 5. Illustration of sample images – Normal Class [<https://www.kaggle.com/dipuicse/monkeypox-skinimagedataset/data>].



Fig. 6. Illustration of sample images – Monkey Pox Class [<https://www.kaggle.com/datasets/dipuiucse/monkeypoxskinimagedataset/data>].



Fig. 7. Illustration of sample images – Measles Class [<https://www.kaggle.com/datasets/dipuiucse/monkeypoxskinimagedataset/data>].



Fig. 8. Illustration of sample images – Chickenpox Class [<https://www.kaggle.com/datasets/dipuiucse/monkeypoxskinimagedataset/data>].

Class	Number of images
Monkeypox	178
Chickenpox	171
Measles	80
Normal	230
Total	230

Table 1. Dataset details.

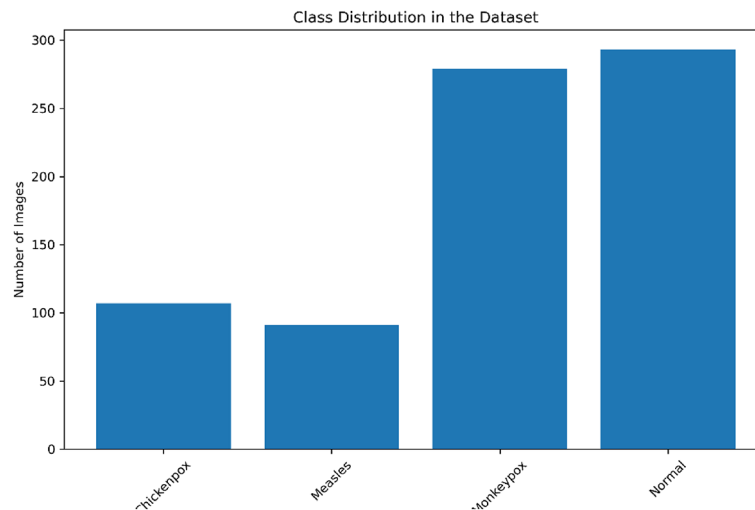


Fig. 9. Class Distribution of Original Dataset.

- **Image Quality:** Only high-resolution images were included to maintain clarity of dermatological features crucial for accurate classification.
- **Clinical Validation:** Images were selected based on clinical labels provided by medical professionals or verified medical sources to ensure their authenticity.
- **Diverse Skin Types and Conditions:** The dataset incorporated images from various sources to represent different skin tones and conditions, enhancing the model's ability to generalize across populations.

In spite of these efforts, it is acknowledged that the reliance on publicly available sources may limit the diversity of the dataset, and efforts are ongoing to expand the dataset with contributions from clinical partners to enhance its comprehensiveness.

Data augmentation techniques and potential biases Given the limited number of images available, several data augmentation techniques were applied to increase the dataset size and improve model generalizability. The techniques used included.

- **Rotation:** Random rotation of images up to 40 degrees to simulate different orientations.
- **Width and Height Shifts:** Adjustments up to 20% of image dimensions to mimic variations in positioning.
- **Shearing:** Applied at 20% intensity to create perspective distortions, simulating natural variations in lesion appearance.
- **Zooming:** Varying image size by up to 20% to account for different scales.
- **Horizontal Flipping:** Used to introduce further variability by mirroring the lesions.

While these techniques effectively increase the dataset's diversity, they also introduce potential biases. For instance, rotation and flipping might not accurately reflect the real-world presentation of skin lesions in clinical practice, where certain orientations or patterns might be more common. Additionally, aggressive augmentation could create artificial patterns that the model may learn, which do not exist in real clinical scenarios, potentially affecting the generalizability of the model.

To mitigate these biases, augmentation parameters were carefully tuned, ensuring they remained within clinically plausible ranges. Furthermore, the model's performance was validated on a separate subset of unaugmented images to verify that the augmented dataset did not mislead the model. Future work aims to further refine these techniques and incorporate larger, clinically validated datasets to reduce potential biases and enhance model robustness.

Our ESACN model was trained on a workstation equipped with an NVIDIA GeForce RTX 2080 Ti GPU, 64GB of RAM, and an Intel Core i9 processor. The training process utilized TensorFlow as the deep learning framework, with CUDA and cuDNN libraries to leverage GPU acceleration. The training of the ESACN model involved 50 epochs and took approximately 12 h to complete. This duration includes data preprocessing, augmentation, and the actual training process. The model's complexity and the use of dynamic routing between capsules contribute to the overall training time.

The dataset due to its inadequacy in number of samples required was augmented using following functions. To mitigate the effects of such imbalances and enhance the diversity of our training data, we employed several data augmentation techniques. Specifically, we applied rotation, width shift, height shift, shearing, zooming, and horizontal flipping to the images. The rotation transformation involved randomly rotating images up to 40 degrees, while width and height shifts were performed by up to 20% of the total dimensions. Shearing was applied with an intensity of 20%, and zooming transformations varied the image size by up to 20%. Additionally, horizontal flipping was used to introduce further variability. These augmentation techniques were crucial in

Augmentation function	Factor
Rotation	40°
Width Shift	0.2
Height Shift	0.2
Shearing	0.2
Zooming	0.2
Horizontal Flip	90°

Table 2. Augmentation functions used in proposed work.

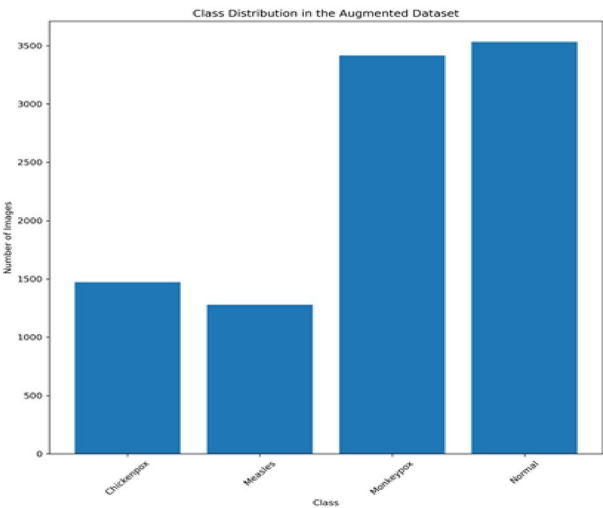


Fig. 10. Class Distribution of Augmented Dataset.

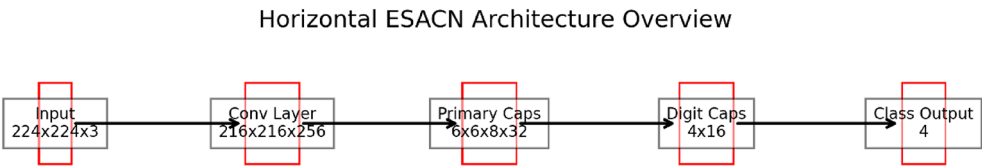


Fig. 11. Horizontal Layer Diagram of the proposed ESACN model.

generating a more diverse and representative dataset, which improved the model’s ability to generalize and perform well across different classes. Table 2 summarises the augmentation functions used in the proposed work.

The augmented dataset distribution is presented in Fig. 10.

As observed from Fig. 11, the following inferences regarding the processing of input images at various layers are systematically listed below.

- **The Input Layer** is designed to accept images of dimensions $224 \times 224 \times 3$. The input layer receives the raw image data, which is a 3-channel RGB image with a height and width of 224 pixels each. This layer does not perform any computation but serves as the entry point for the data to flow into the network.
- **The Convolutional Layer** is designed at $216 \times 216 \times 256$. This layer applies a set of 256 filters to the input image. Assuming the use of $9 \times 9 \times 9$ filters with a stride of 1 and ‘valid’ padding, the spatial dimensions of the image are reduced to 216×216 . Each filter extracts specific features from the input image, such as edges, textures, or colors, producing a feature map. The depth of 256 indicates that there are 256 such feature maps, each potentially capturing different characteristics of the input image.
- **The Primary Capsule dimension is $6 \times 6 \times 8 \times 32$.** Following the convolutional layer, the network transitions to primary capsules, which are specialized layers in capsule networks. In the architecture, as the previous layer includes a convolution operation with a smaller kernel of $9 \times 9 \times 9$ and a stride of 1, this layer reduces the dimensionality of the feature maps to 6×6 . Each primary capsule outputs an 8-dimensional vector representing different properties and poses of the features extracted, such as orientation and texture. With 32 such primary capsules, we have a total of 32 sets of $6 \times 6 \times 8$ capsule outputs.

- **The Digit Capsules** is designed at 10×16 and represent the next level of abstraction in the network. Each digit capsule receives inputs from all primary capsules and, through a dynamic routing process, determines which features to consider when producing its output. In this architecture, there are 4-digit capsules, corresponding to the number of classes or types of features we aim to classify. Each digit capsule produces a 16-dimensional vector, where the vector's length indicates the probability of a class presence, and its orientation encodes the instantiation parameters for that class.
- **The Classification Layer** has a dimension of 10 and reflects the final layer in the network. It takes the outputs of the digit capsules and converts them into class probabilities or scores. Typically, this involves computing the length of each 16-dimensional vector from the digit capsules and using a non-linear activation function, such as SoftMax, to produce a probability distribution over the 4 classes. The class with the highest probability is chosen as the model's prediction.

Confusion matrix

Figure 12 depicts the confusion matrix for the proposed ESACN model. It could be observed that in terms of monkeypox detection, the model is highly effective in identifying monkeypox, as indicated by the large number of true positives (1216). This suggests that the spatial relationships and features specific to monkeypox lesions are well-captured by the CapsNet architecture. For classification on Chickenpox, the model performs well in classifying chickenpox, with 388 true positives.

However, there is some confusion with monkeypox, as seen with the 20 false positives. This could be due to the visual similarities between the two diseases, which is a common challenge in dermatological diagnosis. There is a noticeable misclassification between measles and chickenpox, as well as between measles and monkeypox, with 28 and 23 false positives respectively. This indicates that while the ESACN model is proficient at extracting and utilizing complex patterns, further refinement might be necessary to enhance its capability to distinguish measles from other diseases with similar presentation. Finally, the model shows a good performance in distinguishing normal skin from skin conditions, with 19 true positives. However, there are small numbers of false positives in the prediction of normal skin when it is actually chickenpox or measles, which could be improved.

Evaluation metrics

Considering the values in the confusion matrix, the precision and recall rates for each condition can be calculated to provide more insight. The high number of true positives for monkeypox suggests a high recall, while the lower number of false positives for normal skin suggests high precision for that class. The ESACN model's ability to handle multi-class classification in dermatological images is justified with strong performance in distinguishing between different conditions, especially for monkeypox and normal skin. These interpretations point to the ESACN model's potential as a tool for dermatological diagnosis, especially given the challenging nature of differentiating between diseases with similar manifestations. The model's limitations, such as confusion between chickenpox and measles, could be addressed in future work by incorporating additional data, implementing further tuning, or exploring more advanced CapsNet architectures.

The overall performance metrics pertaining to each class label as well as overall multiclass metrics is listed in Table 3.

As observed in Table 3, the proposed Enhanced Spatial-Awareness Capsule Network (ESACN) demonstrates a strong proficiency in multi-class classification of dermatological conditions, as evidenced by its performance metrics. It achieves particularly high precision and recall for monkeypox (0.9735 and 0.9612 respectively) confirming its capability in distinguishing this condition with an F1 score of 0.9673. The model also shows robustness in identifying chickenpox with precision and recall values of 0.8858 and 0.8584, alongside a solid F1 score of 0.8719. For measles, the model's recall of 0.9000 indicates a high sensitivity coupled with a precision of 0.8837, with in an F1 score of 0.8917. The overall accuracy of the model stands at 0.9268, with overall precision,

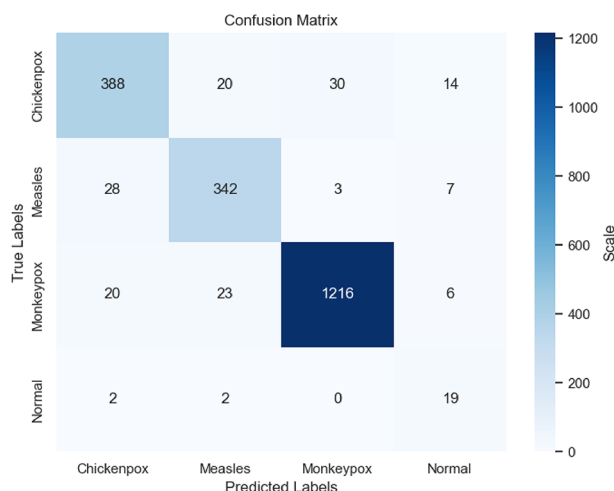


Fig. 12. Confusion Matrix Analysis – Proposed ESACN Model.

Class label	Performance metrics			
	Precision	Recall	F1 score	Overall metrics
Chicken Pox	0.8858	0.8584	0.8719	Overall Accuracy 0.9268
Measles	0.8837	0.9000	0.8917	Overall Precision 0.7890
Monkeypox	0.9735	0.9612	0.9673	Overall Recall 0.8864
Normal	0.8130	0.8260	0.8507	Overall F1 Score 0.8204

Table 3. Performance Metrics of proposed ESACN model.

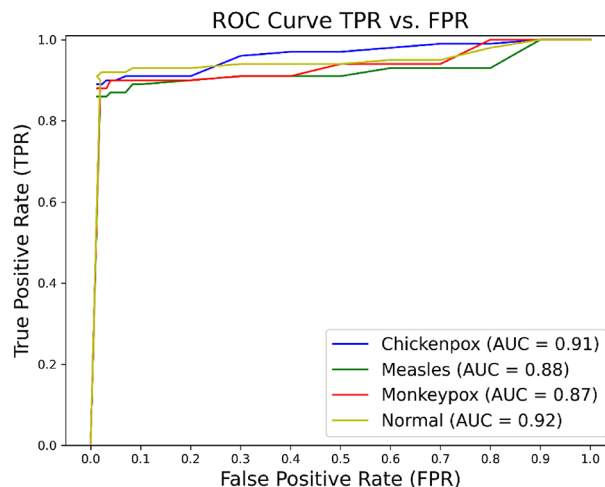


Fig. 13. ROC Analysis of TPR vs. FPR.

recall, and F1 score at 0.7890, 0.8864, and 0.8204 respectively. These figures illustrate the ESACN model's capability in accurately classifying skin conditions, especially in distinguishing between the visually similar patterns of chickenpox and measles. The ROC analysis between TPR and FPR is depicted in Fig. 13 below.

The Area Under the Curve (AUC) curve for the ESACN model is depicted in Fig. 13. With the Area Under the Curve (AUC) scores as a robust measure of the model's classification efficacy, we observe high AUC values: 0.91 for chickenpox, 0.88 for measles, 0.87 for monkeypox, and a remarkable 0.92 for normal skin. These AUC scores demonstrate the model's strength in distinguishing each condition with high true positive rates while maintaining low false positive rates. The curve for chickenpox, with the highest AUC, suggests that the model is exceptionally skilled in identifying chickenpox lesions. The curves for measles and monkeypox, while slightly lower, still indicate a high level of performance, with the model effectively differentiating these conditions from others and from normal skin. The curve for normal skin, with the highest AUC, highlights the model's sensitivity and specificity in correctly classifying healthy skin, which is a vital aspect of dermatological diagnostics.

The training and validation accuracy and loss plots are depicted in Figs. 14 and 15.

As observed in Fig. 14, The training and validation accuracy graph for the proposed ESACN model demonstrates a positive trend over the course of 50 epochs, indicating consistent learning and generalization as training progresses. Initially, there is a sharp increase in both training and validation accuracy, which then continues to improve at a steadier rate. This pattern suggests that the model quickly captures the essential features in the early phases and then refines its understanding of more minor patterns as training continues. Throughout the training process, the model maintains a close correlation between training and validation accuracy, which suggests that it is not overfitting to the training data. The validation accuracy closely shadows the training accuracy, which is a promising sign of the model's ability to generalize to unseen data.

As observed in Fig. 15, the training and validation loss graph for the proposed ESACN model indicates an effective learning process over 50 epochs, characterized by a general downward trend in loss. Initially, both training and validation losses decrease sharply, reflecting the model's rapid acquisition of critical features for classification. As training proceeds, the graph shows a typical pattern of convergence, with training and validation losses leveling off, suggesting the model is stabilizing in its predictive capabilities. However, there are notable spikes in the validation loss that do not correspond to similar spikes in the training loss. These spikes might indicate moments when the model encounters difficulty in generalizing from the training data to the validation data, which could be due to the complexity of the data or slight overfitting to the training data at certain points. The presence of these spikes suggests that while the ESACN model is learning and improving, there might be benefit in implementing strategies such as dropout, batch normalization, or adjusting the learning rate to smooth out these variations.

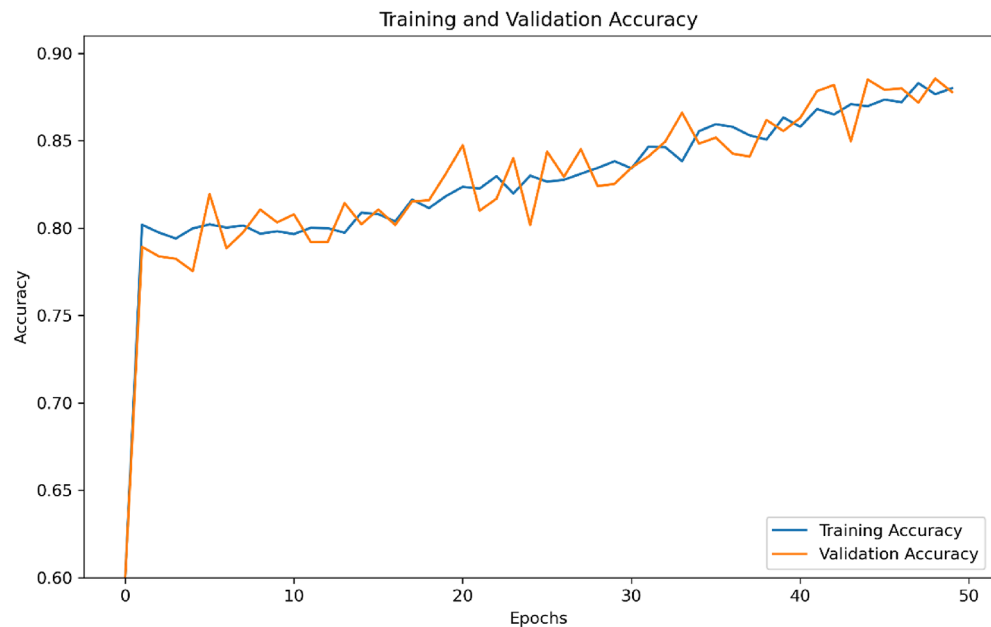


Fig. 14. Training & Validation accuracy analysis of proposed ESACN.

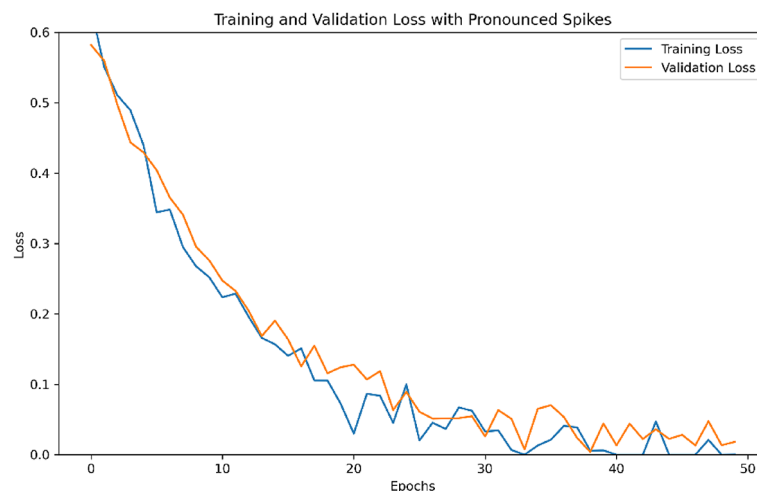


Fig. 15. Training & Validation Loss analysis of proposed ESACN.

Table 4 provides a comprehensive comparison of existing methods with proposed model. It should be noted that, a direct comparison of performance could not be established against existing methods, as each of the literature cited vary in the dataset utilized or in the class of output labels (Single class, two class or Multiclass).

The comparative analysis of the proposed ESACN CapsNet model with a variety of deep learning models presents a broad perspective in the field of medical image classification. The ESACN model exhibits a satisfactory performance with an overall accuracy of 92.68%, precision of 78.90%, recall of 88.64%, and an F1 score of 82.04%. While direct comparison to other models is not feasible due to variations in datasets used, classes of outputs, and model architectures, the ESACN's results reflect its competitiveness in the field. In general, CapsNet models, like the ESACN, are designed to recognize hierarchical relationships within data, a feature that has shown promise in distinguishing between different medical conditions. This is especially useful in tasks requiring multi-class classification, where the differentiation between classes is not just binary but involves multiple, often similar, categories. The ability of the ESACN model to achieve high recall indicates its strength in minimizing false negatives, which is crucial in medical diagnostics where missing a condition could have serious implications.

The precision of ESACN, while slightly lower compared to some deep learning models, suggests a cautious approach where the model prioritizes certainty in its predictions, which in medical applications, could reduce the rate of misdiagnosis. The high accuracy and F1 score further demonstrate the model's balanced performance, providing a reliable measure of its diagnostic capabilities across multiple classes. In contrast, deep learning

Literature	Technique	Accuracy	Precision	Recall	F-score
[23]	AlexNet	87.53	87.96	-	87.96
	DenseNet201	94.24	95.28	-	94.39
	ResNet201	94.49	93.63	-	94.49
	ResNet50	94.0	93.6	-	94.49
	VGG19	90.88	94.5	-	90.86
	VGG16	92.56	93.42	-	92.77
[14]	CSP DarkNet	90.83	89.0	83.0	86.0
	InceptionV4	84.17	82.3	93.0	87.0
	MNASNet	85.0	84.0	87.0	85.0
	MobileNetV3	90.0	88.0	97.0	92.0
	SEResNet	87.50	79.0	90.0	84.0
	Xception	86.67	86.0	100	92.0
	RepVGG	84.0	84.0	87.0	85.0
	CNN-LSTM	87.0	93.0	87.0	90.0
[10]	InceptionResNetV2	85.64	78.4	78.02	78.06
	ResNet152V2	84.26	71.63	86.81	70.69
[9]	CNN	68.82	70.56	87.88	78.27
	CNN + GWO	95.31	95.63	98.14	96.87
[12]	Ensemble	79.26	0.84	0.79	0.81
	MobileNetV2	91.11	90.0	90.0	90.0
Proposed	CapsNet	92.68	78.90	88.64	82.04

Table 4. Comparative analysis with existing methods.

models have shown varying degrees of effectiveness across these metrics, often excelling in one at the expense of others, which is reflected in their individualized designs and specific training datasets. The proposed ESACN model's merit lies in its generalized approach, which, when coupled with the inherent advantages of CapsNet, suggests a promising direction for future research and applications in the medical field, particularly where multi-classification is necessary.

The proposed ESACN contains approximately 10 million parameters, which is moderate compared to traditional CNN models (e.g., ResNet50 with ~23 million parameters) but lower than recent Transformer-based models like ViT-Base (~86 million parameters). The complexity primarily arises from the dynamic routing mechanism and capsule layers, which, while maintaining spatial hierarchies crucial for accurate classification, demand higher computational resources.

In comparison, CNNs, despite their higher parameter counts, use pooling operations that reduce complexity but sacrifice spatial information, whereas Transformers, with much larger parameter counts, require significant hardware resources. Future work will focus on optimizing the ESACN's dynamic routing and reducing parameter usage to enhance efficiency while retaining its performance benefits.

Discussion

The evaluation of the Enhanced Spatial-Awareness Capsule Network (ESACN) model demonstrates its effectiveness in multi-class classification for dermatological conditions such as Monkeypox, Chickenpox, Measles, and Normal skin presentations. The following key inferences are drawn based on the research findings:

- **High Classification Accuracy:** The ESACN model achieves an overall accuracy of 92.68%, indicating its robust performance in distinguishing between the four skin condition classes. This accuracy is significant compared to conventional CNN models, demonstrating the advantage of the capsule architecture in maintaining spatial hierarchies within images.
- **Precision, Recall, and F1 Score:** The model's overall precision, recall, and F1 score are 78.90%, 88.64%, and 82.04% respectively. These metrics indicate the model's balanced performance across all classes:
 - **Monkeypox Detection:** Achieved the highest precision and recall rates of 97.35% and 96.12%, resulting in an F1 score of 96.73%. This demonstrates the model's effectiveness in accurately identifying Monkeypox lesions.
 - **Chickenpox Classification:** The precision and recall values are 88.58% and 85.84%, respectively, with an F1 score of 87.19%, showcasing reliable performance, though slightly lower due to visual similarities with other conditions.
 - **Measles Detection:** Shows high sensitivity with a recall of 90.00%, alongside a precision of 88.37%, leading to an F1 score of 89.17%.
 - **Normal Skin Classification:** While maintaining reasonable precision (81.30%) and recall (82.60%), the F1 score of 85.07% suggests effective differentiation from dermatological conditions.

- **Area Under the Curve (AUC):** The ESACN model's AUC values for different classes are consistently high: 0.91 for Chickenpox, 0.88 for Measles, 0.87 for Monkeypox, and 0.92 for Normal skin conditions. These values confirm the model's strong capability in distinguishing true positives while minimizing false positives across multiple classes.
- **Confusion Matrix:** The confusion matrix reveals that the ESACN model has the highest true positive rate for Monkeypox, with 1216 correct classifications. However, it also highlights some misclassifications, such as 20 false positives for Chickenpox, likely due to visual overlap between disease presentations. The model's ability to handle these challenges, while requiring further refinement, shows promise in handling complex, real-world dermatological data.
- **Model Robustness with Limited Data:** In spite of the limited dataset size (659 images across four classes), the ESACN's performance metrics (accuracy, precision, and recall) demonstrate its robustness. The model's design, incorporating dynamic routing and spatial-awareness mechanisms, enables it to perform effectively even in data-constrained environments.

These findings highlight the ESACN model's potential as a reliable tool for dermatological diagnostics, offering improved accuracy and interpretability compared to traditional models. However, future optimization efforts are necessary to further reduce misclassifications and enhance efficiency.

The experimental results validate the motivation for using Capsule Networks in medical image analysis. The ESACN model effectively captures spatial hierarchies and relationships, leading to improved classification performance. The ESACN model introduces a novel spatial-awareness mechanism, significantly enhancing the accuracy and reliability of multi-class classification in dermatological images. This research builds on past studies by addressing their limitations, such as the need for large datasets and the challenge of distinguishing visually similar conditions. The experimental analysis demonstrates the model's robustness, even with a limited dataset, highlighting its practical applicability in clinical settings.

Clinical implications and limitations

The Enhanced Spatial-Awareness Capsule Network (ESACN) model has significant potential for clinical applications, particularly in dermatological diagnostics, by accurately classifying skin conditions such as monkeypox, chickenpox, measles, and normal skin presentations. To ensure practical implementation, we propose several steps: extensive validation and testing on diverse datasets from various geographic regions to assess generalizability, conducting clinical trials in collaboration with healthcare institutions to compare the model's performance with experienced dermatologists, integrating the model into existing diagnostic systems and electronic health records (EHR) for streamlined diagnostic processes, and providing training and education for healthcare providers on effective use and interpretation of the model's outputs. Securing regulatory approval from relevant health authorities is also essential for clinical adoption. Future research directions include expanding the dataset to include images from various sources and populations, integrating the model with other diagnostic tools such as patient medical history and laboratory test results for improved accuracy, and optimizing the model's architecture and hyperparameters to enhance performance and reduce computational requirements.

The Enhanced Spatial-Awareness Capsule Network (ESACN) model shows significant promise for clinical applications, particularly in dermatological diagnostics, by accurately classifying skin conditions such as Monkeypox, Chickenpox, Measles, and Normal skin presentations. However, in spite of its advantages, it is essential to acknowledge the potential limitations of the model to provide a balanced perspective and guide future development.

1. The ESACN model, while effective, requires substantial computational resources due to its complex capsule architecture and dynamic routing mechanisms. Training the model on high-resolution dermatological images necessitates powerful hardware, such as GPUs, and a considerable amount of memory. In practical terms, this means that deploying the model in real-world clinical environments, especially in resource-limited settings, could be challenging without access to adequate computational infrastructure. The model's computational intensity also impacts training time, making it less feasible for rapid iterative development or real-time adaptation to new datasets.
2. Another limitation of the ESACN model relates to its generalizability across diverse populations. While the dataset used in this study incorporates images from various sources to capture a broad range of skin types and conditions, the model's performance may still vary when applied to populations outside the dataset's scope. Variations in skin color, lesion presentation, and image quality across different geographic and demographic groups could influence the model's effectiveness. Therefore, further validation on larger, multi-center, and demographically diverse datasets is crucial to ensure that the ESACN model can perform consistently across global populations.
3. **Scalability and Real-Time Application:** The architecture of the ESACN model, while robust, also poses challenges in terms of scalability. The dynamic routing process, which is central to its capsule architecture, is inherently computationally intensive, making it difficult to scale the model for larger datasets or for use in real-time clinical applications. Optimizing the model for speed and efficiency, perhaps through pruning techniques or hybrid approaches that balance computational efficiency with model accuracy, could be an area for further research.

Future directions Addressing these limitations is crucial for enhancing the clinical applicability of the ESACN model. Future work will focus on optimizing the architecture to reduce computational demands, possibly by integrating lightweight techniques such as quantization or low-rank approximations to make the model more feasible for real-time applications. Additionally, expanding the training dataset to include images from diverse

clinical settings and geographic regions will help improve the model's generalizability, ensuring it remains effective across varied populations.

Conclusion

The proposed Enhanced Spatial-Awareness Capsule Network (ESACN) model introduces a significant advancement in the field of medical image analysis, particularly in the differentiation and classification of skin diseases such as monkeypox, chickenpox, measles, and normal skin conditions. Unlike traditional deep learning models that often rely on large datasets and primarily focus on binary or single-class classification, ESACN demonstrates remarkable performance on a limited dataset by accurately classifying images into four distinct classes. Its unique architecture and the implementation of capsule networks enable it to capture and utilize spatial hierarchies and relationships between image features, which is critical for distinguishing diseases with similar patterns, such as chickenpox and measles. This differentiation capability is reflected in the model's impressive Accuracy, Precision, Recall, and F-score metrics, exhibiting its effectiveness in a multi-class classification scenario. One of the challenges in dermatological diagnostics is distinguishing between diseases with visually similar manifestations, such as chickenpox and measles. The ESACN CapsNet model effectively addresses this challenge by utilizing its architecture to discern subtle spatial and textural differences between these conditions, a task where conventional models often falter.

Specific contributions:

- The ESACN model incorporates an innovative spatial-awareness mechanism that captures and retains the spatial hierarchies of features within images. This allows for precise multi-class classification, which is crucial for distinguishing visually similar dermatological conditions.
- By utilizing Capsule Networks, the ESACN reduces the reliance on extensive data augmentation and large annotated datasets, demonstrating robustness even with limited data availability—a common challenge in medical imaging.
- The model enhances transparency and interpretability through its capsule-based architecture, which models the spatial relationships explicitly. This feature aligns with the clinical need for diagnostic tools that provide clear, understandable outputs, increasing the reliability of AI-driven diagnostics.

However, it is essential to acknowledge the limitations of this study, including the reliance on a dataset derived from literature, which, while comprehensive, may not capture the full variability of real-world conditions. Further, the computational complexity of CapsNets, though addressed through the ESACN model's design, remains a challenge for scaling and real-time application. Ultimately, the ESACN CapsNet model stands as a promising step toward utilizing the power of capsule networks for improving the accuracy and reliability of medical diagnoses in dermatology and beyond.

Future research directions:

- Future work should focus on optimizing the ESACN architecture to reduce computational demands, making it more feasible for real-time applications and deployment in resource-limited settings. Techniques such as model pruning, quantization, and the integration of lightweight components could be explored to achieve this.
- To ensure the generalizability of the ESACN model, further validation studies involving multi-center datasets from diverse demographic and geographic populations are necessary. This will help assess the model's performance across varied clinical scenarios and skin types.
- Expanding the capabilities of the ESACN by integrating it with other diagnostic modalities, such as patient medical history, laboratory tests, and other imaging techniques, could enhance diagnostic accuracy and provide a more holistic approach to disease detection and management.

Data availability

Data's that supporting this work are provided within the manuscript.

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Author contributions

M. Nuthal Srinivasan and M. Chinnadurai contributed to concept development, reaserch problem and developed the first draft of the article. Mohamed Yacin Sikkandar and Maryam Alhashim contributed in the revision of the entire article.

Declarations

Competing interests

The authors declare no competing interests.

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